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Paleogeography of some vestiges of Iapetus: Paleomagnetism of the Ordovician Robert's Arm, Summerford, and Chanceport Groups, central Newfoundland

ROB VAN DER VOO
REX J. E. JOHNSON
BEN A. VAN DER PLUIJM
LAURA C. KNUTSON*

Department of Geological Sciences, University of Michigan, Ann Arbor, Michigan 48109-1063

ABSTRACT

In the Central Mobile Belt of Newfoundland, three volcanic groups of early Middle Ordovician age (Robert's Arm, Summerford, and Chanceport) are thought to have formed within the Iapetus Ocean. Demagnetization analyses of 328 samples from these three groups has yielded similar magnetic directions, calculated by combining stable endpoint directions and great-circle poles. The three groups yield similar poles located at 6°N, 140°E; 8°N, 140°E; and 10°N, 131°E, corresponding to paleolatitudes of about 30°S. The magnetization of the Robert's Arm Group passes the tilt test, and one site shows normal polarity, antipodal to the directions of the other sites. The paleolatitude of 31°S places the volcanic rocks at the time of their formation in the middle of Iapetus, between the Laurentian margin at about 15°S and the location of the Avalon Terrane at >50°S. Subsequent accretion to Laurentia and the presently adjacent island-arc rocks represented by the Moreton's Harbour Group occurred after northward drift in the later Ordovician/earliest Silurian and before the Middle Silurian.

INTRODUCTION

Newfoundland comprises the classical divisions of an orogenic belt formed in Paleozoic continent-continent collision (Williams, 1979; Williams and Hatcher, 1982, 1983; van der Pluijm and van Staal, 1988). In West Newfoundland, the Humber Zone consists of Grenville-aged basement of the North American craton (Laurentia) and a lower Paleozoic shelf sequence on which ophiolites, such as the Bay of Islands Complex (Stevens, 1970), have been emplaced. In contrast, the eastern part of Newfoundland, the Avalon Peninsula, is the type area of the Avalon terrane with a latest Precambrian (that is, Pan African) basement upon which Cambrian and Ordovician strata were deposited in epicontinental facies (Rast and Skehan, 1983; Krogh and others, 1988). Located between the basement terranes of Laurentia and Avalon is the, at least locally, allochthonous Central Mobile Belt (Colman-Sadd and Swinden, 1984; van der Pluijm, 1987; Stockmal and others, 1990). This belt contains (Cambro-)Ordovician assemblages of ocean floor and island relicts, island-arc remnants, and sedimentary rocks in deep-water facies. Some of these assemblages are unconformably overlain by Silurian epicontinental sedimentary and bimodal volcanic rocks, such as those of the Springdale Group (Chandler and others, 1987).

In the most recent subdivision (Williams and others, 1988), the Central Mobile Belt is separated into two main zones and several subzones: the Dunnage zone, which is composed of the Notre Dame subzone in the northwest and the Exploits subzone in Central Newfoundland, and the Gander zone in the southeast. The Notre Dame and Exploits subzones are separated along the Red Indian Line. In this subdivision, the Summerford Group lies to the south of the Red Indian Line, whereas the Robert's Arm and (it appears) the Chanceport groups are located north of this boundary.

It has long been suspected that the rocks of the Central Mobile Belt formed in a fairly wide Ordovician ocean, named Iapetus (Harland and Gayer, 1972), by analogy with the similar traverse in Great Britain from the Laurentian margin in northern Scotland through the Central Belt (that is, the Southern Uplands, the Lake District, Anglesey) to the Midland Craton of southern Britain and Wales (Dewey, 1969; McKerrow, 1988); the latter area is thought to be correlative to the Avalon terrane (and has, in fact, been called "East Avalonia"; Scotese and McKerrow, 1990).

The Central Mobile Belt in Newfoundland contains some Ordovician units with a characteristic Celtic fauna that differs from the Laurentian and Avalonian faunas (Neuman, 1984), indicating that Iapetus was quite wide. The abundant extrusions of Early to Middle Ordovician age in the Central Mobile Belt represent relicts of various tectonic elements in Iapetus. We have sampled some of these vestiges of Iapetus for a paleomagnetic study in an attempt to determine their paleolatitudinal distribution and to estimate their distance from the North American and Avalonian margins during Ordovician time. An earlier paleomagnetic study (Schwarz and Bostock, 1978) provided a hint of an early Paleozoic magnetization, but "no further conclusions could be drawn" (Bostock, 1988).

Previous paleomagnetic studies in Laurentia and Avalon provide a paleolatitudinal framework within which results from the Iapetus-derived rocks can be interpreted (Van der Voo, 1988). The Ordovician Laurentian margin was situated between 10°S and 20°S (Van der Voo, 1990; Farr and Sprowl, 1990), whereas Avalon is thought to have been situated adjacent to Gondwana in paleolatitudes generally greater than 50°S for Early to Middle Ordovician times (Weisse and others, 1985; Van der Voo, 1988) and at about 40°S during latest Ordovician time (Johnson and Van der Voo, 1990). Similar Ordovician paleolatitudes have been obtained from rocks in the British traverse (Briden and others, 1984; McCabe and Channell, 1990; Torsvik and others, 1990; see also Cocks and Fortey, 1982).

In a separate paleomagnetic study of the Early Ordovician rocks of the Moreton's Harbour Group, we have obtained a paleolatitude of

*Present address: 155 Sharene Lane, no. 314, Walnut Creek, California 94596.

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about 11°S (Johnson and others, 1991), which places this terrane very close to the Ordovician margin of Laurentia. Given the petrological and geochemical island-arc signature of the Moreton's Harbour Group and associated units (Strong and Payne, 1973; Kean and Strong, 1975; Williams and Payne, 1975), and the subduction implied by this volcanism, a model has been constructed (van der Pluijm, and others, 1990) in which the main part of Iapetus Ocean floor is located between this island arc and Avalon/Gondwana to the south, and is being subducted northward under the arc during Ordovician time. In contrast, the negligible separation between the arc and the Laurentian margin implies that a back-arc situation was most probable for that part of Iapetus; this back-arc basin is thought to have closed during the Middle Ordovician, with the collapse of the basin giving rise to the Taconic Orogeny and the emplacement of the Humber Arm and Hare Bay allochthons.

In this study, we will attempt to extend this scenario to include the paleogeography of the ocean-island and arc rocks to the south of the Moreton's Harbour Group. The extrusions of the Robert's Arm, Summerford, and Chanceport Groups are ideal for paleolatitude study, because their stratification planes can be well determined. Thus, inclinations can be measured with respect to the paleohorizontal, and, because inclinations (but not declinations) are invariant to rigid-body rotations of any type, paleolatitude can be determined. The low grade of regional metamorphism (prehnite-pumpellyite facies; Bostock, 1978) precludes a thermal resetting of the remanence, although a chemical remanence acquisition remains a possibility.

GEOLOGY AND SAMPLING

The Robert's Arm, Summerford, and Chanceport Groups are units of similar age, consisting, where sampled, of mostly pillow lavas. The Robert's Arm Group has yielded Middle Ordovician ages (455 Ma, Rb/Sr; 473 ± 2 Ma, U/Pb; Dunning and others, 1987; Bostock, 1988). Limestone blocks with Llanvirn-Llandeilo conodonts have been interpreted (Nelson, 1981) as having been deposited on top of Robert's Arm basalts, supporting the notion that these are no younger than Middle Ordovician (Bostock, 1988). A Llanvirnian age for the Summerford was determined from extensive fossil evidence (Neuman, 1984); the age of the Chanceport Group is based on regional correlation (Arnott and others, 1987). The outcrops of the three groups are located to the south of the major and steeply dipping Lobster Cove–Chanceport fault (Dean and Strong, 1977; Williams and others, 1988). This fault separates them from the Early Ordovician Lush's Bight and Moreton's Harbour Groups. Pronounced faunal (Neuman, 1984), structural, lithological, and/or geochemical contrasts (Williams, 1979; Jacobi and Wasowski, 1985; Bostock, 1988; Swinden and others, 1989) have been suggested for various places along this fault.

Regional location maps and schematic geological maps with sampling sites are presented in Figures 1 and 2. Good outcrops of pillow lavas of the Robert's Arm Group are located mostly along the coasts of the inland bays (Fig. 1). Bedding tops and stratification planes are clearly indicated by the pillow structures in our sampling sites, so that strikes and dips, which are generally steep to overturned, are unambiguous (Bostock, 1978, 1988). Most flows face west to north, but oppositely facing sections are also present (see also Bostock, 1978, 1988), and they have been sampled for a tilt test. All of our sites are in pillow basalt, with the exception of site 15, which is in massive basalt. We have not sampled any of the more felsic units nor the less abundant sedimentary rocks.

In the outcrop area of the Robert's Arm and Lush's Bight Groups, an overlap assemblage is formed by the late Early Silurian (429 +6, –5 Ma; late Llandovery) Springdale Group (Chandler and others, 1987), indicating that juxtaposition of the Ordovician formations, an early phase of deformation and, presumably, accretion to the North American margin, preceded Middle Silurian times (Bostock, 1978; Chandler and others, 1987). Although the angular unconformity between the Springdale and Robert's Arm Groups attests to a pre–Middle Silurian

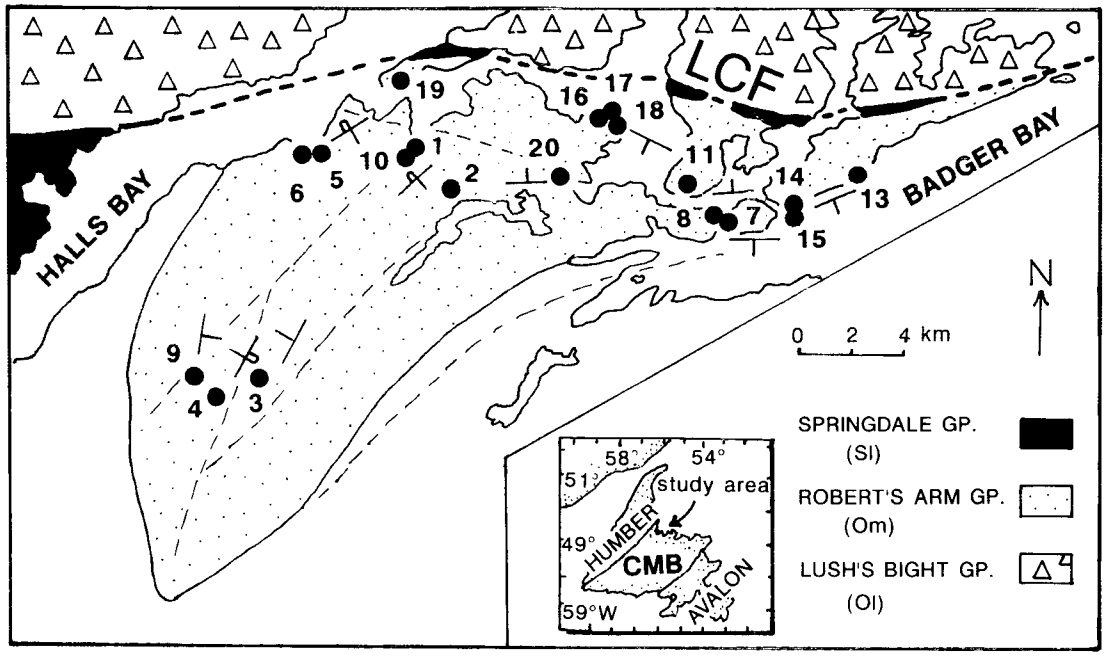


Figure 1. Schematic geological map of the outcrop area (after Bostock, 1988) of the Robert's Arm Group, with sampling sites and general bedding attitudes. The Lower Silurian (Llandovery) Springdale Group forms an overlap assemblage on top of the Ordovician formations (Chandler and others, 1987). LCF = Lobster Cove fault; CMB = Central Mobile Belt.

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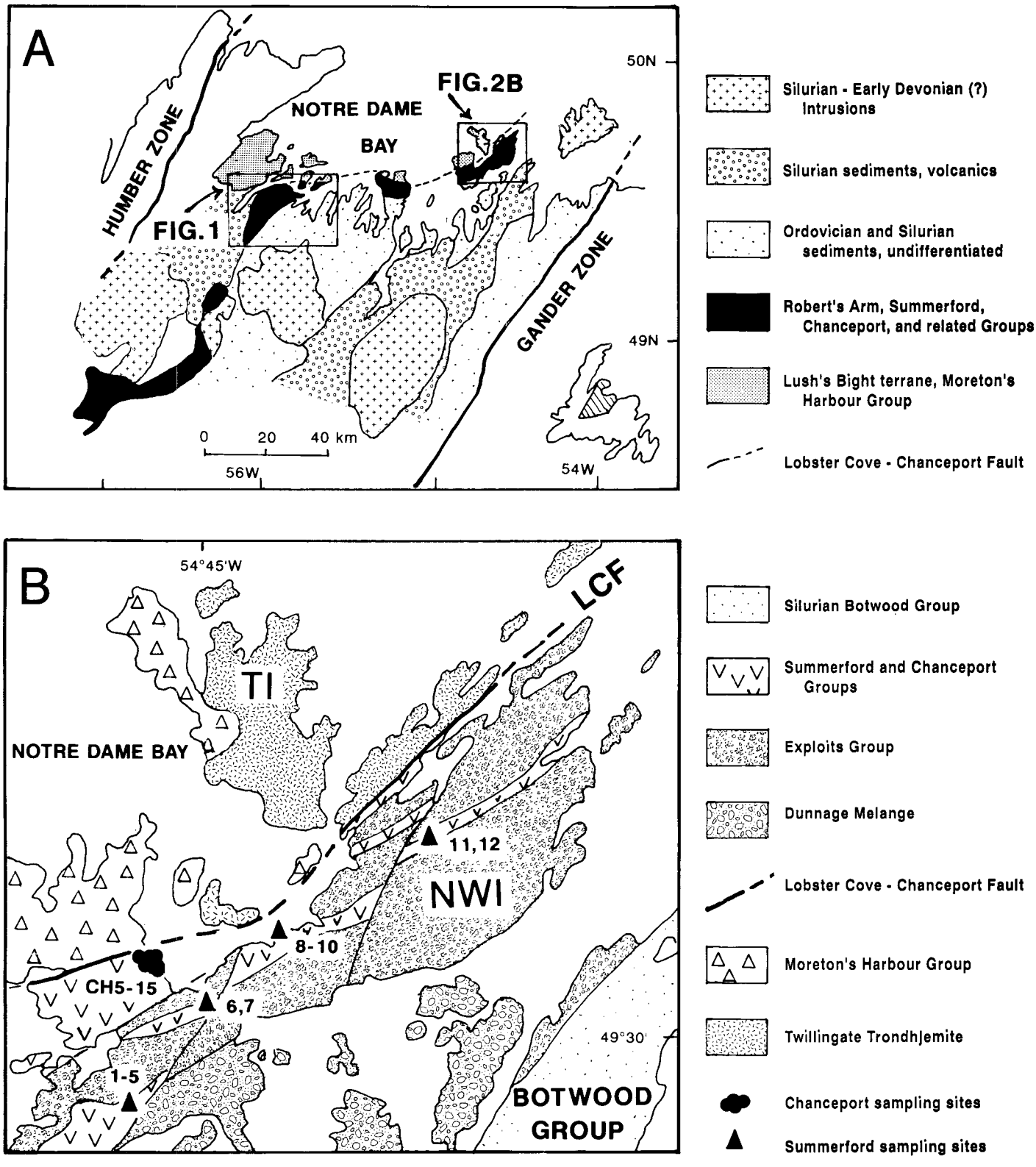


Figure 2. (A) Simplified regional map of the Notre Dame Bay area (after Bostock, 1988) with the locations of the maps of Figure 1 and Figure 2B. (B) Schematic geological map of the outcrop area of the Summerford and Chanceport (CH) Groups, with sampling sites. NWI = New World Island, TI = Twillingate Island, LCF = Lobster Cove–Chanceport fault.

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deformation, the Springdale Group itself is also folded. Thus at least two deformational events, one Middle Ordovician–Early Silurian (Taconic, or early Caledonian/Acadian) and the other Late Silurian–Devonian (Caledonian/Acadian), affected the Robert's Arm Group.

In the more eastern part of Notre Dame Bay (Fig. 2), rocks of the Summerford, Chanceport, and Cottrell's Cove (not sampled) groups form an east-northeast–west-southwest–trending elongate belt between the Moreton's Harbour Group to the north of the Lobster Cove–Chanceport fault and the Dunnage Mélange and correlatives to the south. The overlap assemblage here is the Middle Silurian Botwood Group (van der Pluijm and others, 1991), although it does not directly overlie the Summerford and Chanceport Groups.

In the northern and western part of the Robert's Arm outcrop area, where we sampled, the lavas appear fresh, although Bostock (1978) reported microscopic observations of pervasive early spilitization. Although the rocks are dissected by many faults and were strongly tilted during the accretion process of the Late Ordovician to Early Silurian, the lavas are not cleaved, nor do they appear to have undergone penetrative deformation at the outcrop scale.

Sampling was carried out with a portable, gasoline-powered drilling apparatus in the summers of 1987–1989. Orientation of the samples was done by Brunton compass and clinometer; intensities of magnetization are too low to influence the measurements. A total of 328 samples was demagnetized from the collections for which we report here the paleomagnetic results: 166 samples (19 sites) from the Robert's Arm, 59 (9 sites) from the Summerford, and 103 samples (11 sites) from the Chanceport Groups. Each flow is called a site; for the Robert's Arm Group, sites are widely distributed, whereas for the Chanceport and some of the Summerford sites, collecting was done in a single stratiagraphic section. The thicknesses sampled are difficult to estimate because of possible structural repetition by thrusting; they are at least 100 m for the Robert's Arm Group and at least 30 m for the Chanceport and Summerford Groups.

PALEOMAGNETIC ANALYSIS

Samples were measured in the paleomagnetic laboratory of the University of Michigan with a ScT two-axis cryogenic magnetometer or a Schonstedt spinner magnetometer. Alternating-field demagnetization with Schonstedt apparatus of pilot specimens proved to be of limited value, because the components of magnetization could not be separated due to overlapping coercivities. Thus, the entire collection of samples reported on was treated thermally with Schonstedt apparatus in typically more than 14 demagnetization steps.

Demagnetization results were plotted in orthogonal demagnetization diagrams (Zijderveld, 1967) and in stereographic projections for visual inspection; directions were determined by principal component analysis (Kirschvink, 1980). Two or more components of magnetization were present in almost all samples, and in many cases these components had overlapping blocking temperatures, causing a continuous migration of measured directions in stereonets. It has been possible, however, for a large majority of such samples to fit great-circle (planar) trends to the trajectories, indicating that only two components were simultaneously being removed. Without the use of the sophisticated great-circle analysis developed by Halls (1976), and by him and his colleagues subsequently (for example, Bailey and Halls, 1984), this paper would be able to list ancient paleomagnetic directions for only 35% of the entire collection, whereas with the combined use of stable end-point directions and great circle poles, we are now able to report characteristic directions for about 80% of the total number of samples.

Robert's Arm Group

Intensities of the natural remanent magnetizations (NRM's) ranged from 0.001 to a little more than 1 A/m. NRM directions streaked generally between the present-day geomagnetic field direction and upward directions in the southern quadrants, before correction for the tectonic tilt.

In Figure 3, demagnetization diagrams for six samples (from five different sites) show a relatively simple behavior: after removal in the first 100° of a random (laboratory induced?) component, two other components of magnetization are successively revealed. The first, generally removed between 100 and 450 °C, conforms to the present-day geomagnetic direction in Newfoundland before correction for tilt; for the samples illustrated in Figure 3, its direction after tilt correction is generally northerly to northwesterly and upward. The second component is generally removed between 450° and 590°, but occasionally also in higher temperatures up to 630 °C, and it reveals a southerly to southeasterly downward direction after tilt correction. In this paper, we call this the "characteristic direction." Site 17 shows antipodal north and upward directions after tilt correction (Fig. 3), whereas in *in situ* coordinates, the directions are upward to the west, which is very different from that of the present-day field. Characteristic directions decaying more or less linearly to the origin, as shown by the samples illustrated in Figure 3, are typical for about 70 of the 166 samples analyzed; site-mean directions are listed in Table 1, with one column (Nd) giving the number of samples that showed such "stable end point" behavior. The characteristic directions before tilt correction are more scattered than after correction; we will return to this issue below when we discuss the tilt test.

A more complicated behavior is shown by the samples illustrated in Figure 4, which all came from the same site. The demagnetization diagrams are continuously curved, and inspection of the changes in direction during demagnetization shows that no stable end points are reached (see the stereographic plots of samples ORG1.H and B in Fig. 4). This illustrates that in the demagnetization of these samples, two (or more) magnetization components are removed simultaneously. Where a great circle can be fitted to the trajectories of the stereonets (in the examples shown, above 150 °C), one can be reasonably certain that there are no more than two components that make up the remanence. Three or more components would not generally lie in the same plane, with the result that the directional trajectory in the stereonet cannot be fitted by a great circle: an example of the simultaneous removal of three (or more) components, to be discussed below, is shown in Figure 5D.

Several sites had only a few samples for which stable end-point directions could be determined, whereas other samples yielded only great circles. Table 1 lists the number of the latter samples in the Ng column. Mean directions were calculated by combining the Nd and Ng results following the method of Bailey and Halls (1984). No sample result has been included more than once in the site means: for example, if a sample yielded a stable end-point direction as well as a great circle pole, only the former was used.

One site (6) for which no stable end points were obtained shows distributions of great circles with well-defined intersections (such as those of site 1 in Fig. 4). The normals (or poles) to these great circles (sample GC's) in such a case will themselves be distributed along a great-circle trend, that is, the site great circle (site GC). The best approximation of the common intersection of all sample great circles corresponds to the site GC pole, defined by the normal to this site GC, which conforms to the characteristic direction. For site 6, it was thus possible to estimate the characteristic direction, even though not a single stable end-point direction could be determined for this site (see Table 1).

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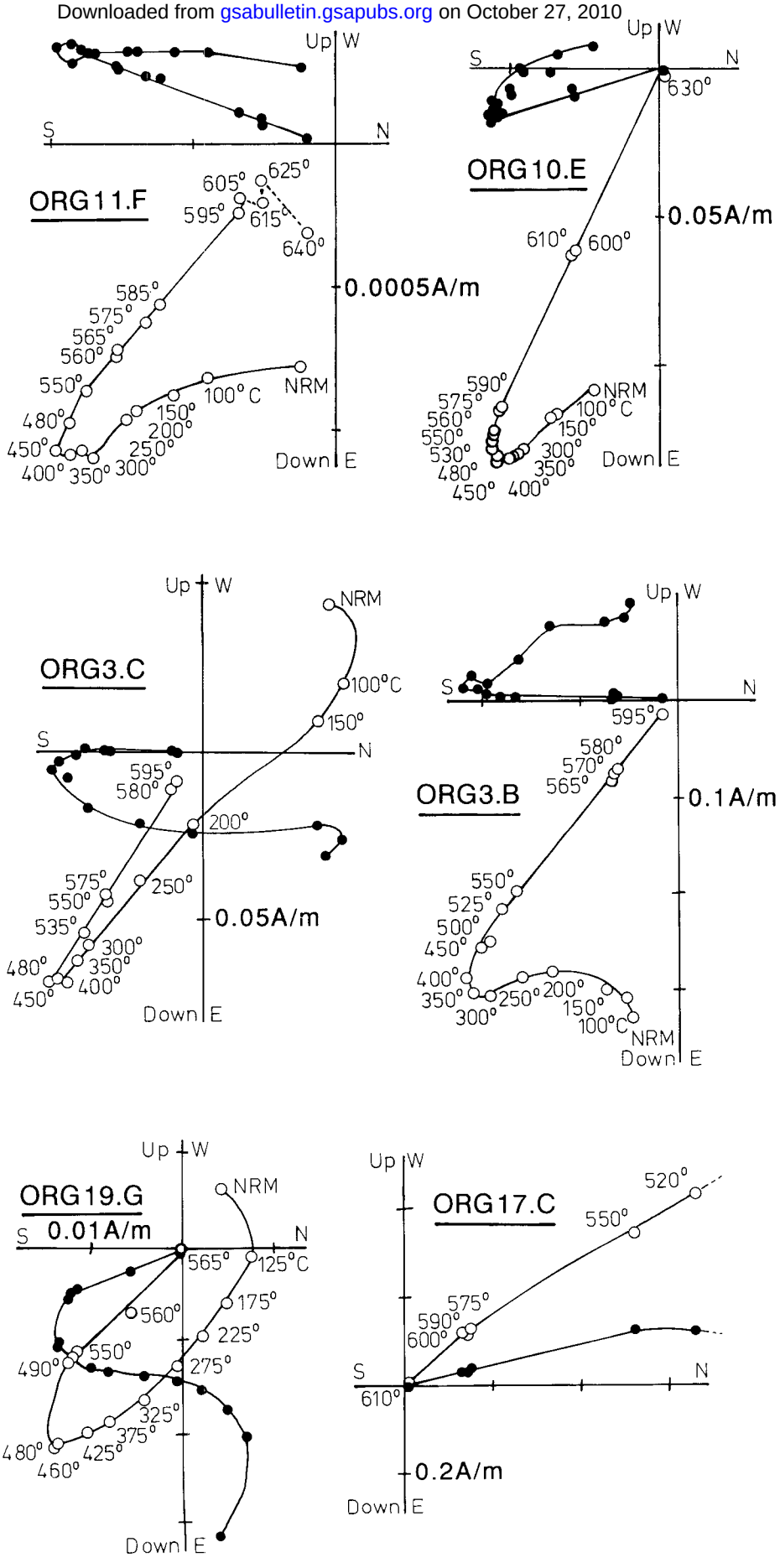


Figure 3. Demagnetization diagrams (Zijderveld, 1967) of typical samples from five different sites of the Robert's Arm Group that yield characteristic directions at higher temperatures, shown after correction for tectonic tilt. Closed (open) symbols represent projections onto the horizontal (vertical) plane; intensities of magnetization are labeled along the axes. The lower-temperature components (150–480 °C) correspond to the present-day geomagnetic field direction in *in situ* coordinates. Sample ORG 17.C shows a normal-polarity magnetization at higher temperatures.

For five other sites for which no stable end points were revealed, great circles were mostly parallel, such that no site-GC trend and, hence, no common intersection could be determined from them (sites 4, 5, 9, 13, and 14). For these sites, however, we have calculated Fisherian averages of the sample GC poles and listed these, marked by an asterisk, in Table 1. In the final averaging of all site means, these five (site-mean) poles can be included according to the method of Bailey and Halls (1984).

Forty-five samples have not been used in the site means of Table 1, for reasons that are worth illustrating briefly in Figure 5, so that a full accounting is given of all results. All samples from site 15 and several isolated samples from other sites revealed only the present-day geomagnetic field direction (Fig. 5A). In some samples, a sudden drop of the magnetic intensity to the origin occurred during demagnetization at high temperatures (Fig. 5B), without defining a great circle. As already noted above, some samples could not be used because they contained three or more overlapping components of magnetization (Figs. 5C, 5D). Some samples had somewhat noisier behavior at high temperatures (for example, Fig. 5E and Fig. 5A above 560°), hampering the fitting of great circles. Finally, one site (16) showed exclusively a southeasterly shallow component of magnetization (Fig. 5F), which after tilt correction does not resemble the characteristic direction. We interpret this magnetization to be of late Paleozoic (Kiaman) age, similar to those found in several other units from Newfoundland (Irving and Strong, 1984, 1985). A hint of the same late Paleozoic overprint is occasionally seen in other samples (for example, Fig. 5E?), but because of overlapping blocking temperature spectra or somewhat noisy behavior, the directions could not be determined precisely.

In summary, 121 of the total of 166 samples, and 16 of 19 sites yield characteristic directions or corresponding great circle poles. In Figures

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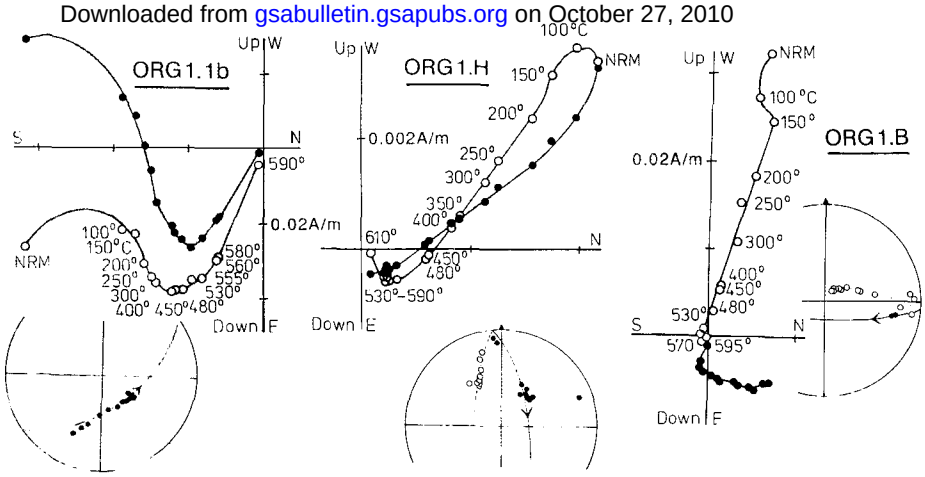


Figure 4. Demagnetization diagrams after tilt correction (as in Fig. 3) of samples from site 1 of the Robert's Arm Group which do not give characteristic ("stable end-point") directions, but which yield demagnetization trajectories in stereographic projections that converge upon the characteristic direction. In the stereographic plots, closed (open) symbols represent projections onto the lower (upper) hemisphere.

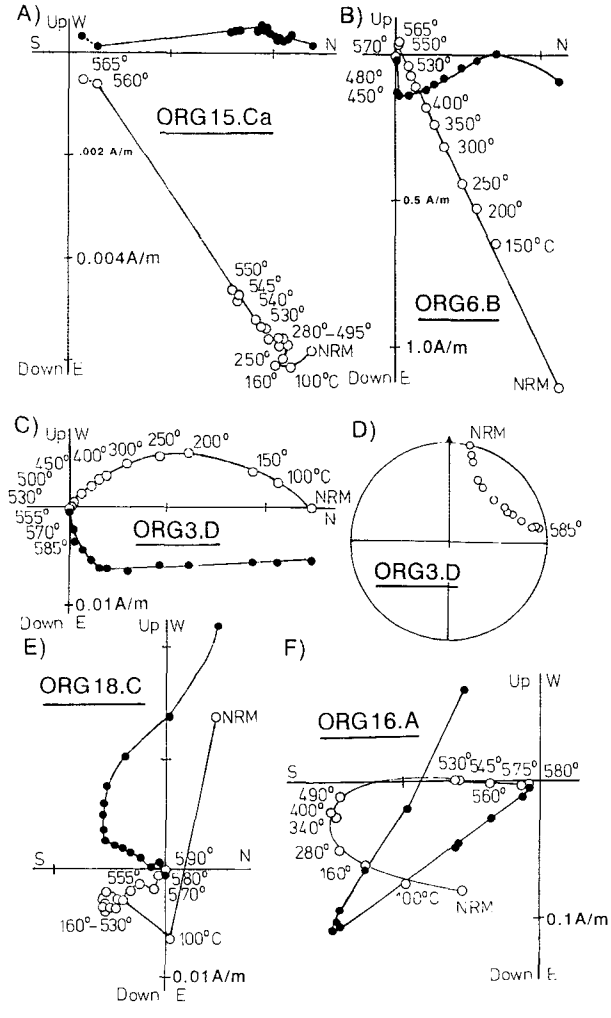


Figure 5. Demagnetization diagrams (as in Fig. 3) in *in situ* coordinates of samples of the Robert's Arm Group which do not yield characteristic directions, nor converging great-circle trajectories, and which have not been used in the final compilation. (A) Sample from site 15 which reveals only the present-day geomagnetic field direction. (B) Sample from site 6 with present-day field direction and (above 565 °C) a sudden drop of intensity to the origin. (C) Sample from site 3 which reveals three different but overlapping components of magnetization, as shown by the stereographic plot of the trajectory of (D), which does not conform to a great circle. (E) Sample from site 18 which shows a noisy decay to the origin such that no great-circle trajectory can be adequately fitted to the directions. (F) Sample typical of those from site 16 which shows a well-defined southeasterly shallow direction of magnetization, which appears to be a late Paleozoic magnetic overprint; upon correction for tilt, this direction becomes southeasterly and upward, which is very different from the characteristic directions of other sites.

6A and 6B, the site-mean directions and GC poles are plotted before and after tilt correction. *In situ* directions show considerably greater scatter than the ones in bedding coordinates, and improvement in the precision parameter k (from 5 to 12) is statistically significant at the 99% level. Thus the Robert's Arm results pass the tilt test. Stepwise partial correction for tilt gives no indication of a synfolding magnetization.

A more detailed analysis of the structure of the outcrop area of the Robert's Arm Group is also possible, given that from southwest to northeast there is a continuous change in strike from north-northeast to east and back to northeast (Fig. 1). The declinations of the characteristic directions give some indications that the trends were originally straighter, but this is by no means definitive. Because such an analysis of oroclinal bending is rather interpretive, we defer speculations about vertical-axis rotations until the discussion section.

Summerford and Chanceport Groups

NRM intensities, directions, and behavior during demagnetization of the Summerford and Chanceport samples are comparable to those of the Robert's Arm Group. Figure 7 illustrates the stable end-point directions, and Figure 8 shows the great-circle trends of typical Summerford samples. The present-day field overprint is ubiquitous, although a few samples contain only the south/downward characteristic component (for example, OSG12.I; Fig. 7). The magnetization in this sample appears to be carried by hematite as indicated by blocking temperatures up to 670 °C. All Summerford sites yield characteristic directions (Table 1), even those for which no stable end points could be determined, because the sample GC poles are well distributed along a site GC trend, and hence they allow the calculation of a reliable intersection (Fig. 8; Table 1).

Figure 9 illustrates the stable end-point directions, and Figure 10 shows the great-circle trends of typical Chanceport Group samples. The present-day field overprint is always present, albeit in variable proportions (Fig. 9). Note that in tilt-corrected coordinates this component is north-northwest and upward, as was the case for the Summerford and several of the Robert's Arm sites. Again many samples show overlapping blocking temperatures of the two components, such that only great-circle trends could be determined (for example, several samples from site 7 in Fig. 10). Ten Chanceport sites yield characteristic directions (Table 1), including those without stable end points, because the sample GC poles are well distributed along a site GC trend, allowing the calculation of a common intersection.

The site mean and sample directions and great-circle poles of the Summerford and

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TABLE 1. SITE-MEAN PALEOMAGNETIC DATA

Site	Strike/ dip	Nd/Ng/Nt	D/I or GC* before TC	D/I or GC* after TC	D/I or GC* after TC+SC	k	α_{95}
Robert's Arm Group							
1	226/125	3/ 7/10	120/–72	128/+52	144/+52	47	9
2	224/ 95	9/ 3/12	135/–47	135/+48	153/+48	92	5
3	211/ 70	8/ 4/13	148/– 9	166/+50	197/+50	30	9
4	24/ 95	0/ 5/ 8	15/+ 8*	33/+ 9*	71/+ 9*	22	17
5	224/ 95	0/ 1/ 7	148/+56*	305/+28*	323/+28*	..	..
6	221/ 80	0/ 4/ 9	158/–24	168/+47	189/+47	60	44
7	94/ 75	10/ 0/14	46/+76	175/+26	143/+26	61	6
8	264/ 80	8/ 0/ 9	200/+ 2	248/+63	226/+63	83	6
9	10/ 85	0/11/12	15/+ 9*	199/+ 4*	251/+ 4*	79	5
10	226/125	7/ 0/ 7	150/–57	156/+66	172/+66	52	8
11	266/105	6/ 4/10	188/–57	186/+47	162/+47	42	9
13	70/ 70	0/ 4/ 5	72/+ 1*	252/+ 1*	244/+ 1*	19	22
14	70/ 90	0/ 8/ 8	250/+10*	240/ 0*	232/ 0*	83	6
15	70/ 90	0/ 0/ 8					
16	120/ 65	0/ 0/ 6					
17†	120/ 65	4/ 0/ 5	265/–56†	353/–40†	295/–40†	182	7
18	120/ 65	0/ 0/ 8					
19	260/110	9/ 0/ 9	137/–62	150/+42	132/+42	151	4
20	272/ 90	4/ 2/ 6	220/–39	220/+38	190/+38	27	15
Mean before TC		11/ 5/19	167/–56			5	21
Mean after TC		11/ 5/19		165/+55		12	13
Mean after SC		11/ 5/19			161/+52	14	11
Summerford Group							
1	220/105	3/ 5/ 8	150/–37	165/+62		12	22
3	220/105	0/ 4/ 4	181/–56	161/+34		208	21
4	220/105	0/ 7/10	164/–56	155/+41		99	31
6	229/123	3/ 6/11	178/–69	159/+48		24	14
7	229/123	0/ 8/ 8	180/–76	152/+43		90	13
8	251/132	1/ 3/ 5	166/–82	162/+50		462	8
10	251/132	1/ 3/ 4	196/–68	185/+58		35	31
11	248/135	3/ 0/ 4	128/–65	129/+64		14	34
12	248/135	4/ 0/ 5	252/–69	186/+40		34	16
Mean before TC		9/ 0/ 9	172/–68			19	12
Mean after TC		9/ 0/ 9		162/+50		31	9
Chanceport Group							
5	245/112	2/ 5/ 7	167/–70	162/+44		113	9
6	245/112	8/ 0/ 9	185/–58	180/+50		78	6
7	245/112	5/ 4/ 9	179/–57	176/+51		69	7
8	245/117	0/10/10	187/–65	174/+47		17	20
9	243/110	0/10/11	209/–55	188/+35		44	13
10	243/110	2/ 6/ 9	159/–65	157/+45		28	15
11	248/110	0/ 0/ 9					
12	248/110	5/ 3/ 8	144/–56	145/+53		60	8
13	255/110	7/ 0/ 8	180/–58	177/+50		36	10
14	255/110	2/10/12	170/–65	168/+45		72	7
15	248/110	1/ 9/11	179/–42	192/+62		58	14
Mean before TC		10/ 0/11	177/–60			46	7
Mean after TC		10/ 0/11		172/+49		50	7

Note: site-mean paleomagnetic data before and after tilt correction (TC) and correction for strike deviation (SC) in the case of the Robert's Arm Group (as explained in the text), for all sites investigated for this study. Strike and dip are listed, following the left-hand rule for the strike direction; overturned bedding dips are indicated by values larger than 90 degrees. Nd = number of samples (sites) yielding characteristic ("stable end-point") directions; Ng = number of samples (sites) yielding great-circle poles, and Nt = total number of samples (sites) demagnetized. D/I = Declination/Inclination and k and α_{95} are the statistical parameters associated with the means (Fisher, 1953).

Asterisks distinguish site-mean great-circle poles (see text for discussion) from characteristic directions.

†indicates the only normal-polarity site mean of site 17, which has been inverted in the calculation of the overall group mean directions (see also Table 2).

Chanceport Groups are shown in Figure 11, both after tilt correction. Both groups show good clustering of characteristic directions, and the great-circle poles are distributed in broad bands along the overall great circles that define the group mean directions (Fig. 11, top diagrams).

Because the bedding orientations of the sites for both groups are all very similar, a tilt test for the Summerford and Chanceport Groups was not possible. We note, however, that the statistical parameters associated with the Summerford Group mean direction improve upon tectonic correction (Table 1), although this is not statistically significant at the 95% confidence level.

DISCUSSION

Well-defined magnetizations, with nearly identical behavior and very similar mean directions (Table 2), have been observed in the three groups studied. In general, blocking temperatures indicate that the magnetization is probably carried by magnetite, but hematite is also observed. The magnetization of the Robert's Arm Group passes the tilt test. The magnetizations are generally of reversed polarity, but in one site of the Robert's Arm Group, normal-polarity directions have been observed.

In any modern paleomagnetic study, a major concern is whether the observed characteristic magnetization is possibly of secondary origin (for example, Van der Voo, 1989). All rock types have at times yielded such remagnetizations, even in situations where the causes may not *a priori* be evident. Because the characteristic directions of this study pass the tilt test, we can be reasonably certain that the magnetizations are pre–Early Devonian. It could even be argued that they are pre–Early Silurian, but the effects of the two deformation phases seen by the rocks are difficult to separate, given the limited outcrop area of the Llandoverian Springdale Group. The presence of reversals in itself cannot rule out remagnetization. Thus, our only recourse in mustering support for a primary, or at least an Ordovician, age of the magnetization is to discuss our paleopole for the Robert's Arm, Summerford, and Chanceport Groups, located at 8°N, 137°E, in terms of a comparison with reference paleopoles for the early and middle Paleozoic. Our paleopole is very different from the North American reference pole path for all of the Paleozoic (Van der Voo, 1990), and we can thus be reasonably certain that the Robert's Arm, Summerford, and Chanceport Groups rocks did not acquire their magnetization in a position adjacent to the North American craton. In other words, the rocks acquired their magnetization before they accreted.

From geological as well as paleomagnetic considerations (Chandler and others, 1987; van der Pluijm and van Staal, 1988; Gales and others, 1989), it seems very likely that the Central Mobile Belt belonged to North America from Middle Silurian time onward. The magnetization must therefore be older than Middle Silurian. We shall assume that it dates back more or less to the time of formation or early (sea-floor) spilitization of the basalts.

The Robert's Arm outcrop area, as has already been noted, shows curved strike trends. In his structural analyses, Bostock (1978, 1988) made a compelling argument that this curvature was imposed on the Robert's Arm rocks during dextral strike-slip or transpressional emplacement against the Lobster Cove fault. Because movement along the fault at least partially postdates deposition of the Springdale Group, and is thus probably Late Silurian to Early Devonian in age, such oroclinal bending should have affected the declinations of the characteristic directions of the rocks. Figure 12 shows a plot of declination versus strike, and it is apparent that a strongly elongate distribution is present, superposed on the normal, Fisherian, scatter of paleomagnetic results. The elongation is, however, primarily caused by sites 4 and 9, on the one hand, from the north-striking area in the southwest (Fig. 1) and by sites 8 and 20, on the other hand, from the east-striking area in the north. In this analysis, we have included declinations for the five great circle poles (sites indicated by a triangle in Fig. 12). These declinations are inferred by assuming that the characteristic direction has an inclination of 55° (that is, the mean inclination after tilt correction); in so doing, a great-circle plane can be fixed by two constraints, one being the site GC pole (triangle in Fig. 6B) through which it must pass, and the other being that the plane must dip 35° (the

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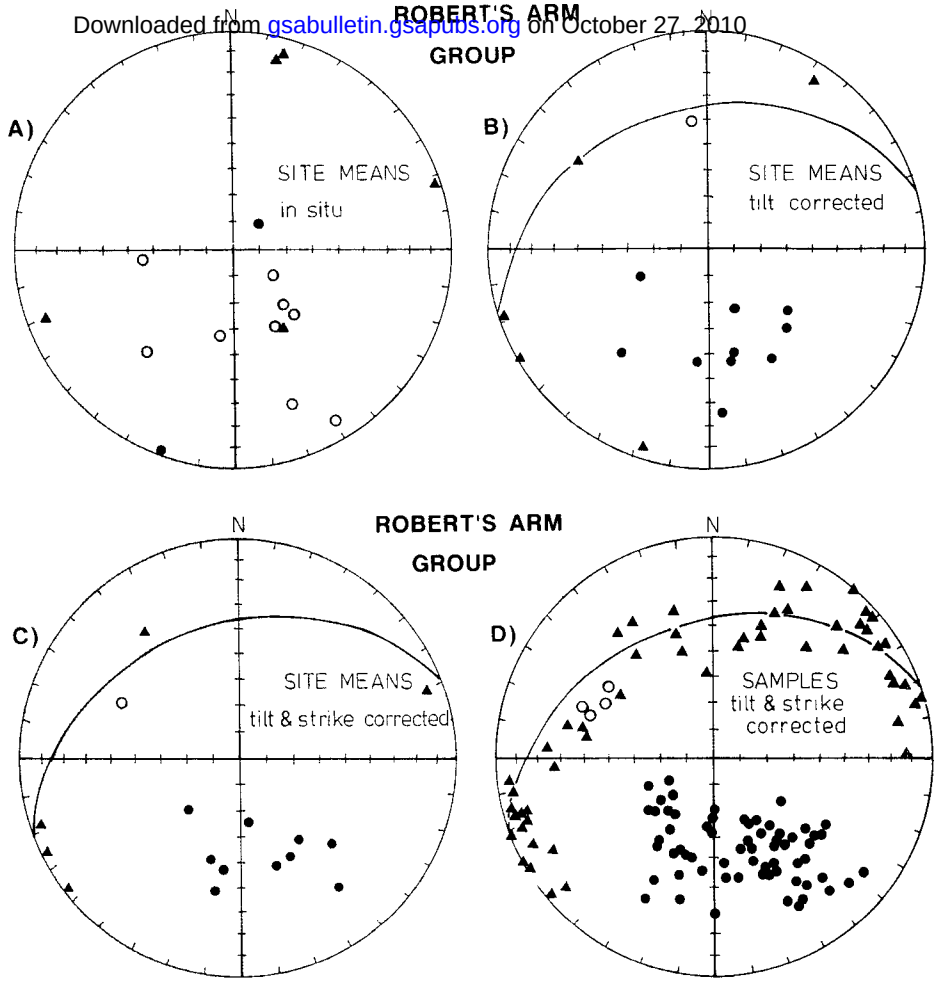


Figure 6. Equal-area plots of characteristic site-mean and sample directions of magnetization (circles) and great-circle poles (triangles) for the Robert's Arm Group as determined from demagnetization trajectories such as those shown in Figures 3 and 4. Closed (open) symbols represent projections onto the lower (upper) hemisphere. (A) Site-mean directions and mean great-circle poles in *in situ* coordinates, showing considerable scatter. (B) Site means in tilt-corrected coordinates. Note the one normal-polarity direction from site 17. (C) Site-mean directions and great-circle poles after correction for tilt as well as for strike trends (see text of discussion section for explanation). (D) Sample directions and great-circle poles after tilt and strike correction.

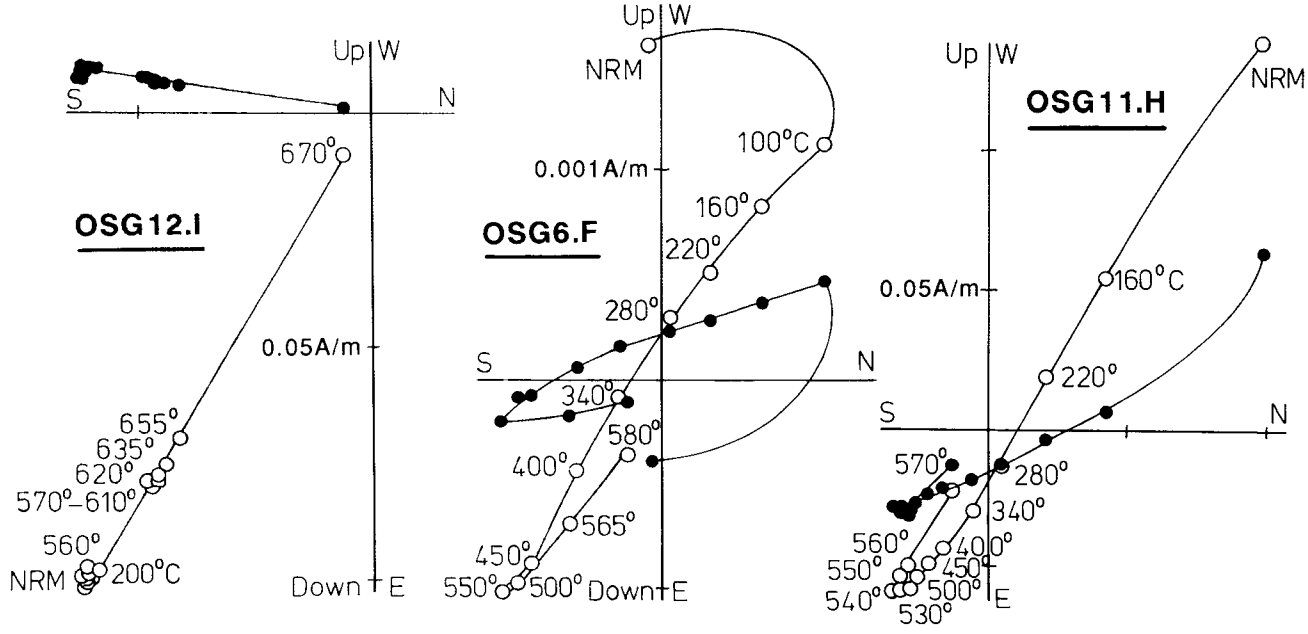


Figure 7. Demagnetization diagrams (as in Fig. 3) of typical samples from three different sites of the Summerford Group that yield characteristic directions at higher temperatures, shown after correction for tectonic tilt. The lower-temperature (100–550 °C) components of magnetization conform to the present-day geomagnetic field direction before tilt correction.

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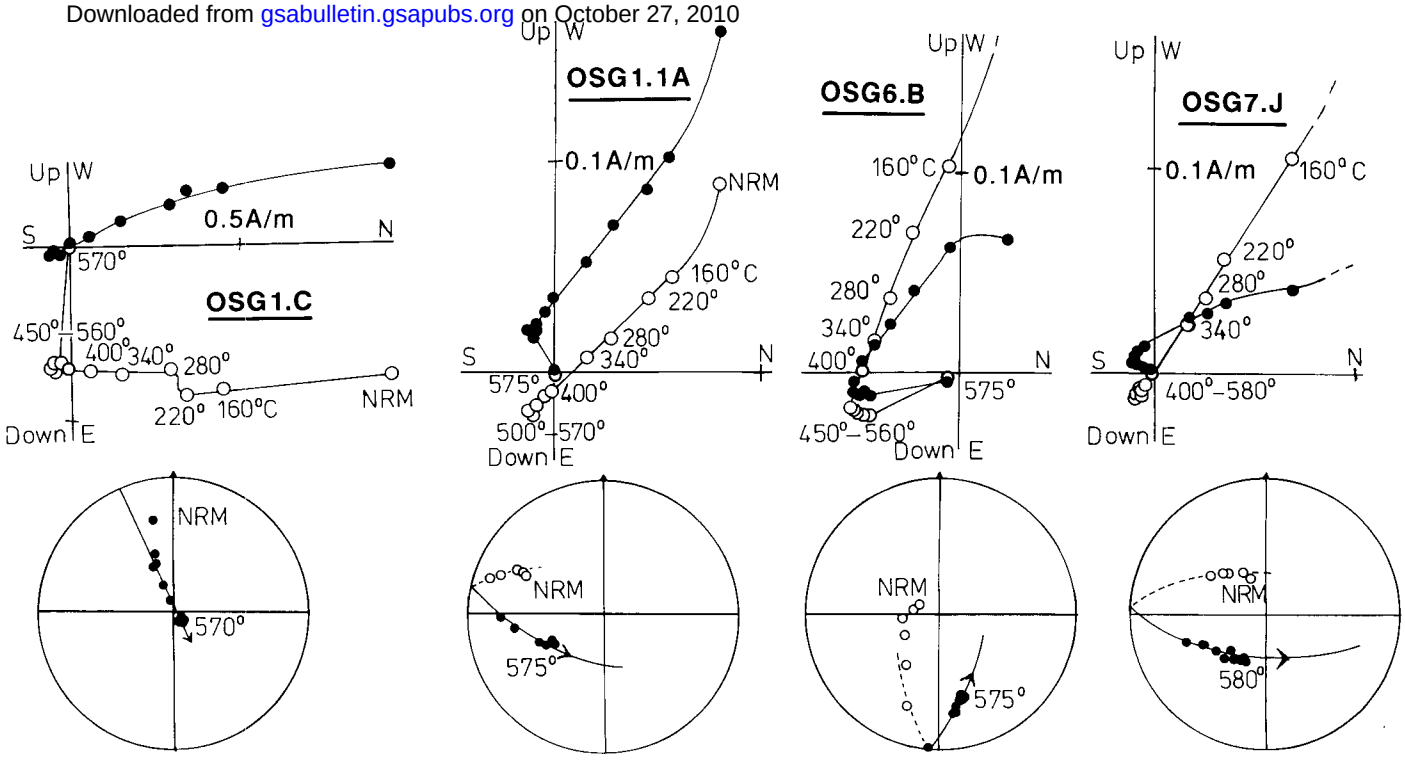


Figure 8. Demagnetization diagrams after tilt correction (as in Fig. 3) of samples from three different sites of the Summerford Group which do not give characteristic ("stable end-point") directions, but which yield demagnetization trajectories (and great-circle poles) that converge upon the characteristic direction as illustrated by the stereographic plots (as in Fig. 4).

complement of 55°). The direction of the normal (pole) to this great-circle plane is the declination plotted in Figure 12. Needless to say that if a site GC pole has near-zero inclination (for example, sites 13 and 14 in Table 1), then the declination is automatically fixed without the dip assumption.

Given the elongate distribution of Figure 12, we have carried out a strike correction, which rotates all sites around a vertical axis by the amount of strike deviation. A site with strike 46°

TABLE 2. PALEOMAGNETIC POLE POSITIONS

Group	Mean D/I after TC, SC	k (K)	α_{95} (A_{95})	Paleopole position
Robert's Arm	161/+52	14	11	6°N, 140°E
Summerford	162/+50	31	9	8°N, 140°E
Chanceport	172/+49	50	7	10°N, 131°E
Mean of 3 Groups	165/+50	374	6	
Mean of 3 Paleopoles		(223)	(8)	8°N 137°E

Note: paleomagnetic pole positions calculated from the mean directions of the three groups investigated in this study, and overall mean direction and paleopole obtained from the three groups combined. Abbreviations as in Table 1.

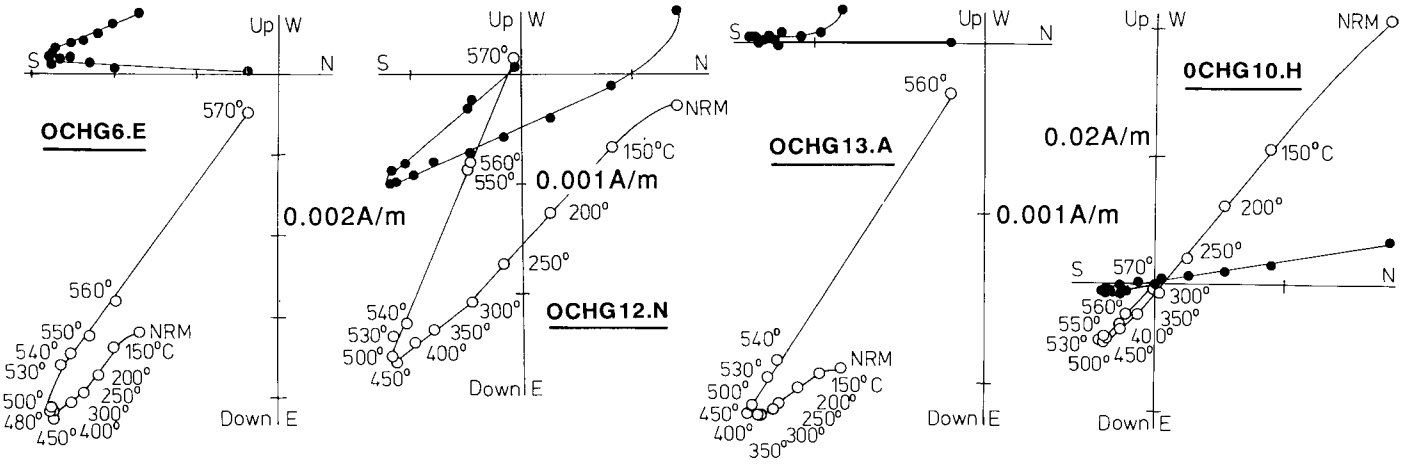


Figure 9. Demagnetization diagrams (as in Fig. 3) of typical samples from four different sites of the Chanceport Group that yield characteristic directions at higher temperatures, shown after correction for tectonic tilt. The lower-temperature (150–500 °C) components of magnetization conform to the present-day geomagnetic field direction before tilt correction.

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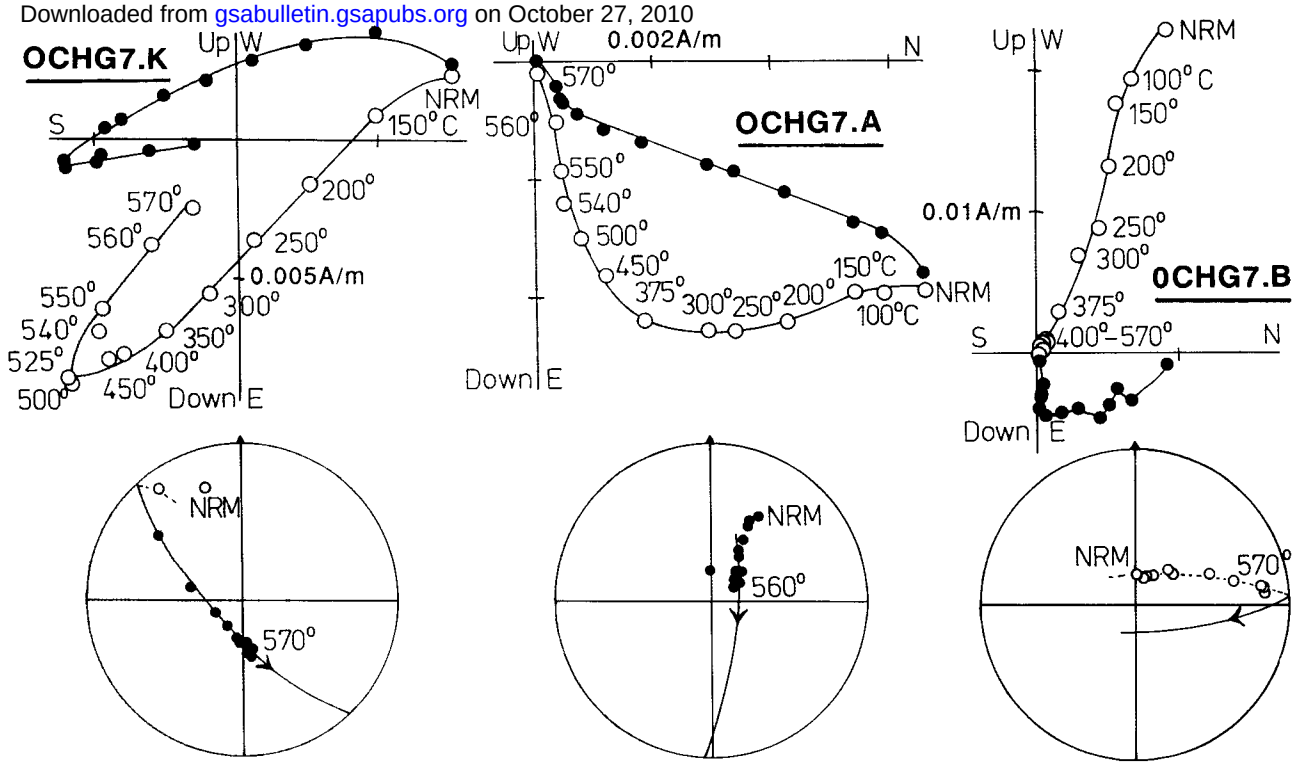


Figure 10. Demagnetization diagrams after tilt correction (as in Fig. 3) of samples from site 7 of the Chanceport Group which do not give characteristic ("stable end-point") directions, but which yield demagnetization trajectories (and great-circle poles) that converge upon the characteristic direction as illustrated by the stereographic plots (as in Fig. 4).

or 226°, for example, deviates by 16° counterclockwise from the regional average of 62° (or 242°); the declination of such a site (for example, site 1 in Table 1) is then corrected from the observed value of 128° to 144°, thereby restoring the inferred rotation.

The third column with site-mean directions in Table 1 gives the strike-corrected declination values, with the inclinations of course remaining unchanged. The overall mean direction for the Robert's Arm Group changes very little, but the statistical parameters improve somewhat; this improvement, however, is not statistically significant (k changing from 12 to 14 for N = 16 entries). In Figures 6C and 6D, the strike-corrected directions are plotted, so that they can be compared with only tilt-corrected directions (Fig. 6B); an improvement can be noted in some of the GC poles and in slightly better clustering of the site-mean directions (dots). On the other hand, correction of the normal-polarity direction of site 17 results in a less antipodal position. In Figure 13, the declinations before strike correction are plotted in map view, together with the generalized strike trends. Again it can be noted that sites 4, 9, 8, and 20 show the clearest correlation with possible oroclinal bending and that sites 7 and 17 show the poorest correlation.

Thus, it is not altogether clear that the oroclinal bending advocated by Bostock and suggested by the strike patterns is strongly

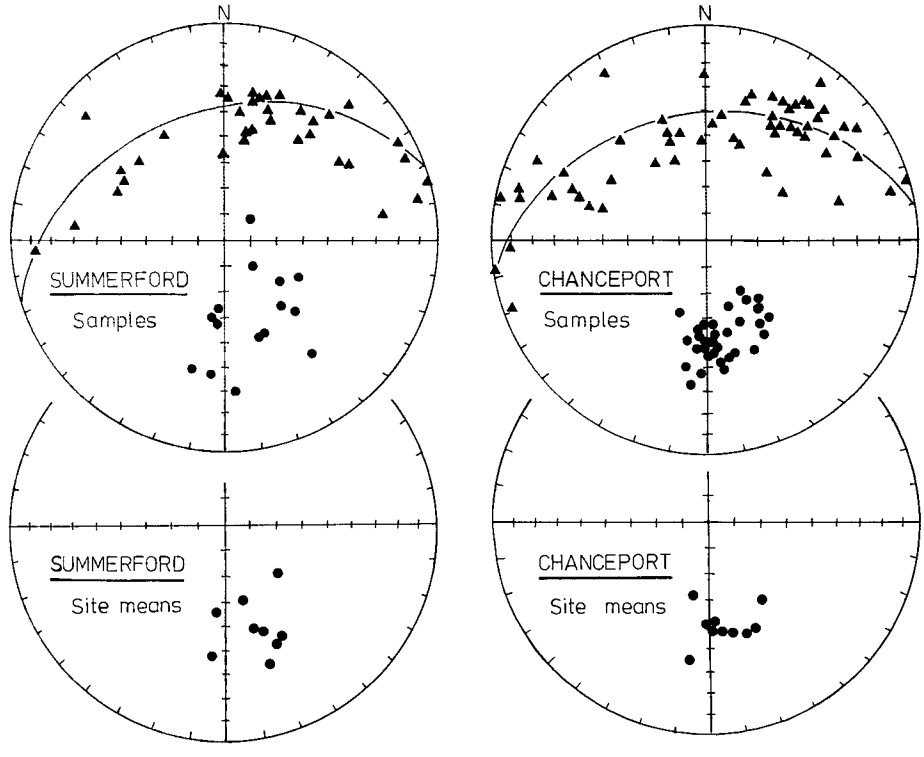


Figure 11. Equal-area plots (as in Fig. 6) of the site-mean and sample directions and great-circle poles after tilt correction for the Summerford and Chanceport Groups.

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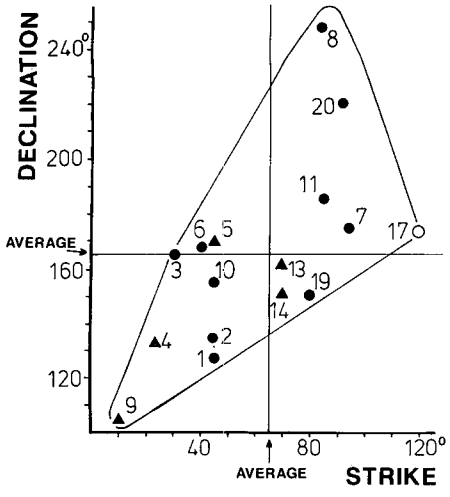


Figure 12. Diagram showing the mean declinations and the strikes of all sites from the Robert's Arm Group which yield characteristic directions (after tilt correction). The elongate diagonal distribution suggests that rotations about vertical axes may have occurred after the acquisition of the characteristic remanence (see also Fig. 13). Symbols correspond to those of Figure 6; for a description of the conversion from great-circle poles to declinations, see the text.

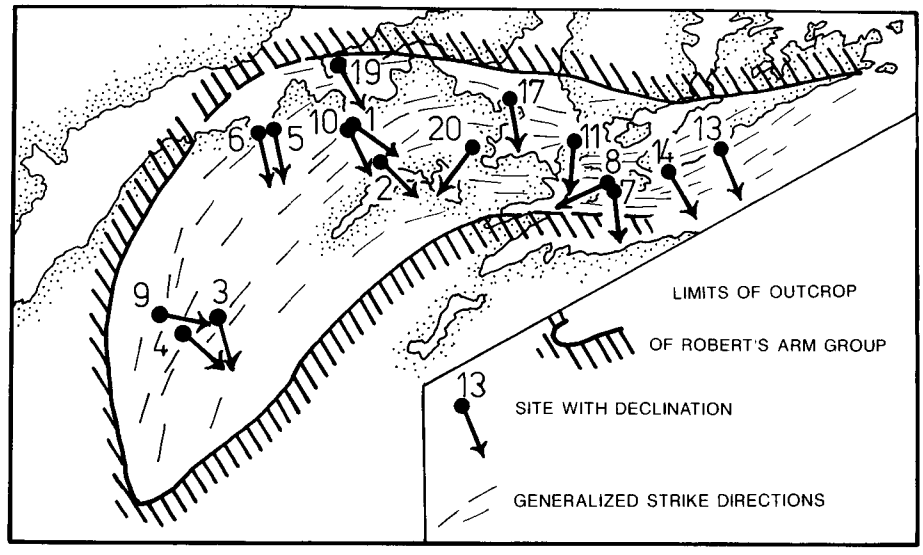


Figure 13. Map of the Robert's Arm Group outcrop area showing the declinations (after tilt correction) of the characteristic site-mean directions (see also Fig. 12). Generalized strike trends are also shown.

supported by the paleomagnetic results. Nevertheless, the overall group mean direction does not change appreciably, and we have used the strike- and tilt-corrected mean in Table 2.

When the three group mean results are combined, we obtain a statistically well defined overall mean direction (D/I = 165°/50°, with k = 374) and a paleopole at 8°N, 137°E, with K = 223 and A_{95} = 8.3°. The corresponding paleolatitude is 31°, which is assumed to be a southerly one following the general consensus about paleolocations of Laurentia, Gondwana, Avalon, and other Appalachian displaced terranes (for example, Scotese and McKerrow, 1990).

The paleomagnetic results presented in this paper allow us to clarify some of the Ordovician paleogeography of Iapetus. Figure 14 illustrates the Middle Ordovician locations of the Iapetus-bordering continents of Laurentia and Gondwana (with adjacent Avalon and Armorica). The paleopositions inferred from the Robert's Arm, Summerford, and Chanceport results are indicated in this figure by a band at 31 ± 8°S. The orthogonal separation of this band from Laurentia's western Newfoundland margin is ~1,500 km, which indicates the minimum oceanic distance these rocks traveled prior to Middle Silurian accretion. Assuming the age of magnetization is ca. 470 Ma, this implies a minimum velocity of about 4 cm/yr. A similar orthogonal distance is seen with respect to Avalon/Armorica (Fig. 14). Convergence between the Laurentian margin and the Robert's Arm, Summerford, and Chanceport rocks may have been oblique or accretion may have occurred at a different location, followed by sinistral or dextral strike-slip (that is, margin-parallel) movement. At this time, it is impossible to speculate about the exact mode of accretion. In this context, we note that the field evidence for the sense of strike-slip movement remains inconclusive; recent workers in Newfoundland have argued for several generations of strike-slip movement (for example, Blewett and Pickering, 1988; Currie and Piasecki, 1989; Keppie, 1989; Lafrance, 1989).

Our results for the Robert's Arm, Chanceport, and Summerford Groups require a modification of the location of the Red Indian Line boundary as proposed by Williams and others (1988), who combined the Robert's Arm Group with the Lush's Bight and Moreton's Harbour Groups into the Notre Dame subzone. The Moreton's Harbour Group yielded a paleolatitude of 11°S (Johnson and others, 1991), dis-

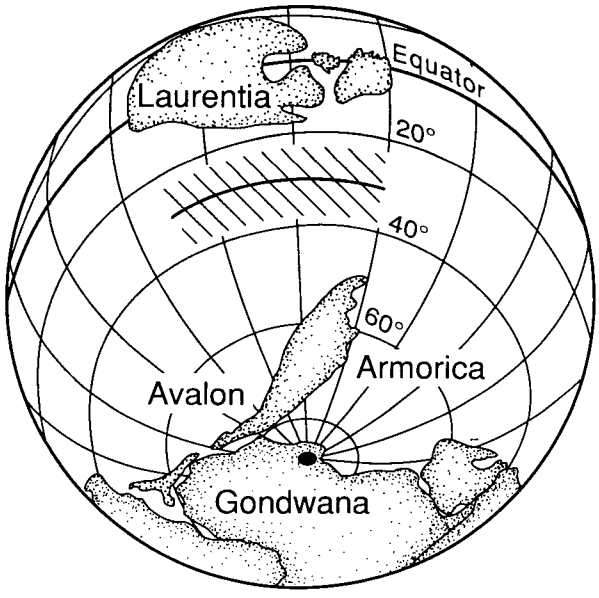


Figure 14. Simplified paleogeographic map showing the early Middle Ordovician positions of the North American craton and Gondwana (with Avalon and Armorica adjacent to it) and the possible paleolatitudinal positions of some of the vestiges of Iapetus (Robert's Arm, Summerford, and Chanceport Groups) as determined by the paleomagnetic inclinations of this study (shaded band at 31° ± 8°S). The map is an interpolation between those of Van der Voo (1988) for the Middle to Late Ordovician and of Johnson and others (1991) for the Early Ordovician.

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tinctly different from that obtained for the Robert's Arm Group (31°S). In contrast, the Summerford Group (placed in the Exploits subzone by Williams and others, 1988) gives the same higher paleolatitude. Thus, we propose that the boundary between the Notre Dame and Exploits subzones be located to the north of the Robert's Arm Group, that is, along the Lobster Cove–Chanceport fault (Fig. 2A).

The early Middle Ordovician paleogeography of Figure 14 agrees with faunal evidence (compare fig. 4 of Cocks and Fortey, 1982, with Fig. 14 in this paper). Neuman (1984), using benthic shelly fauna, concluded that several ocean islands were present in Iapetus. Late Arenig–early Llanvirn fossil assemblages of the Summerford Group, in particular, led to the recognition of the Celtic Province, which was located at intermediate paleolatitudes. An ocean-island setting for the Summerford Group was further suggested from geochemical data (Jacobi and Wasowski, 1985). Unfortunately, faunal evidence for the Robert's Arm and Chanceport Groups is not available. Geochemical evidence for the Robert's Arm Group (Swinden, 1987; Bostock, 1988) suggests an island-arc origin for this unit. The paleomagnetic results of this paper support and quantify the occurrence and position of major volcanic activity in Iapetus. Volcanic rocks of the Robert's Arm, Summerford, and Chanceport Groups were magnetized when they were far distant from both Laurentia and Gondwana and did not accrete to the Laurentian margin until Early Silurian times. This northward motion and subsequent accretion of the belt are recorded in the late Middle Ordovician replacement of the Celtic fauna by the Scoto-Appalachian fauna of the Laurentian margin at more tropical paleolatitudes (Neuman, 1984), and by structural and other paleomagnetic evidence (van der Pluijm and others, 1990).

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